1. INTRODUCTION TO BIOLOGY

Definition

- The word biology is derived from the Greek words, ***bios***, meaning **life,** and ***logos***, meaning **knowledge.**
- Therefore**, Biology** is the branch of science that deals with the study of living **organisms.**
- **Science** is the knowledge about the structure and behaviour of the natural world based on facts that can be approved by experiments.

Branches of Biology

- There are **three main branches** of biology namely:

1. **Zoology-** This is the study of animals. A scientist specialized in this area is called **zoologist.**
2. **Botany-** This is the study of plants. A scientist is called **botanist.**
3. **Microbiology-** This is the study of microscopic organisms. A scientist is called **microbiologist.**

Other branches of biology

1. **Anatomy-** This is the study of the internal structure of living things. A scientist is called **anatomist.**
2. **Physiology-** This is the study of body functions. A scientist is called **physiologist.**
3. **Genetics-** This is the study of inheritance and variation. A scientist is called **genetist.**
4. **Ecology-** This is the study of the relationship between organisms and their environment/ study of living organisms and their surrounding. A scientist is called **ecologist**.
5. **Parasitology-** This is the study of parasites. A scientist is called **parastologist.**
6. **Entomology-** This is the study of insects. A scientist is called **entomologist.**
7. **Cytology-** This is the study of the cell. A scientist is called **cytologist.**
8. **Pathology-** This is the study of diseases. A scientist is called **pathologist.**
9. **Biochemistry-** This is the study of chemical changes inside the organism. A scientist is called **biochemist.**

10. **Morphology-** This is the study of external structure of organisms. The scientist is called **Morphologist.**
11. **Bacteriology-** study of bacteria. The scientist is called **bacteriologist.**
12. **Histology-** study of structure of tissues. The scientist is called **histologist.**
13. **Virology-** study of viruses. The scientist is called **virologist.**
14. **Ornithology-** Study of birds. The scientist is called **ornithologist.**
15. **Ichthyology-** study of fish. The scientist is called **Ichthyologist.**
16. **Taxonomy-** This is the study of classification of organisms. The scientist is called **Taxonomist.**
17. **Embryology-** study of development of organisms from egg to adult. The scientist is called **Embryologist**

Importance of studying Biology.

1. It helps us to understand the developmental stages in human body.
2. It helps in solving environmental problems e.g. ***pollution, shortage of food, global warming / drought, poor health, misuse of natural resources (forests, wildlife, water and soil.***
3. It enables one to pursue careers e.g. ***agriculture, veterinary, public health, medicine, tourism, pharmacy, dentistry, nursing, biology education/ teaching, and horticulture.***
4. It helps us to promote international cooperation in areas like medicine and environmental conservation to solving emerging problems like ***HIV and AIDS.***
5. It helps learner to acquire scientific skills e.g. ***observation , identification, drawing, recording, classifying, measuring, analyzing and evaluating data*** and apply them in daily life.

CHARACTERISTICS OF LIVING ORGANISMS

1. **Nutrition-** a process by which living things acquire and utilize nutrients. Plants synthesize/make their own food using ***light energy, carbon (IV) oxide, water and mineral salts,*** while animals feed on already manufactured foods.
2. **Respiration-** a process in which organic compounds are broken down to produce energy. Energy is used by the organisms to carry out essential activities e.g. growth and movement.

- During respiration, oxygen is usually used while ***energy, carbon (IV) oxide and water*** are the products.
- Living organisms carry out respiration.

3. **Gaseous exchange-** this is the process whereby respiratory gases (oxygen and carbon (IV) oxide) pass across the respiratory surface.

- Examples of respiratory surfaces include **stomata** in leaves, **Alveoli** in lungs, **gills** in fish, **skin** in frogs, **cell membrane** in unicellular organisms.
- Living organisms carry out gaseous exchange.

4. **Excretion-** is the process by which metabolic wastes are separated and eliminated from the body cells. This is to avoid accumulation to toxic levels leading to death.

- Living organisms carry out excretion/ excrete.

5. **Growth and Development-** **Growth** is an irreversible/permanent increase in size and mass while **Development** is the irreversible change in the complexity of the structure of living things.

- Living things grow in order to attain the maximum size and mass which are essential for their body function.
- Living organisms grow and develop.

6. **Reproduction -** is the process by which living organisms give rise to new individuals of the same kind.

- Living organisms reproduce.

7. **Irritability/sensitivity-** This is the ability of living organisms to perceive/detect changes in their surroundings and respond appropriately .

- For example living things react to changes in **temperature, humidity, light, pressure and to presence or absence of certain chemicals.**

6. **Movement**- is a change in position by either a part of or the whole living organism. Living organisms move.

- Movement from one place to another is called **locomotion.**
- Movement in animals include **swimming, walking, running, flying e.t.c.**
- Movement in plants include **closing of leaves, folding of leaves, closing of flowers and growing of shoots towards light.**

Study questions.

1. A car/aeroplane is able to move from place to place and give out exhaust gases however it is not classified as a living organism. Explain.

This is because it does not reproduce, respond to changes in the environment, grow, develop, excrete, carry out nutrition and respire.

2. How does nutrition differ in plants and animals?

Plants manufacture their own food/ are autotrphic; while animals do not manufacture their own food/ are heterotrophic;

3. State the characteristics of living organisms that are specific to plants.

i. ***Autotrophic/ manufacture their own food/ photosynthesis;***

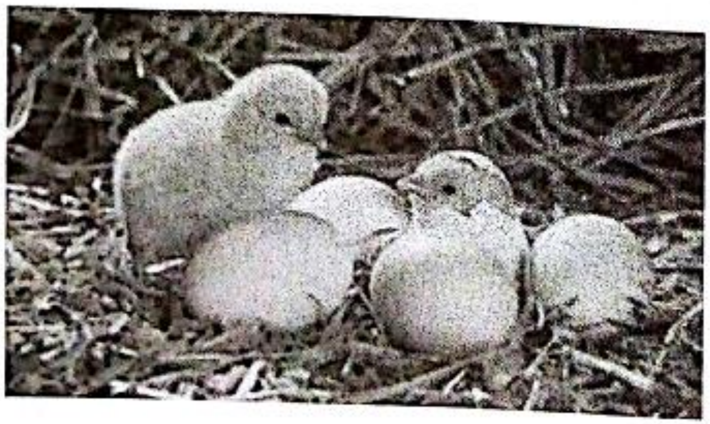
ii. ***Show alternation of generation;***

iii. ***Have limited movement;***

iv. ***Have limited excretory products/ unspecialized respiratory structures;***

v. ***Have localized growth/ growth occurs at specific regions;***

4. The photograph below illustrates living organisms. Study it and answer the question that follow.



- State two characteristics of living organisms illustrated in the photograph

1. ***Reproduction***
2. ***Growth and development.***

5. Name the characteristic of living organisms illustrated by each of the activities described below;

a) Dressing heavily-

✓ **Irritability/ sensitivity/ response to a stimulus**

b) Bursting of the sporangium in *Rhizopus sp.*

✓ Reproduction.

Collection of specimens

- A specimen is a whole organism or part of an organism being studied or examined.

Importance of collection of specimens

- It is important for further study, observations and preservation for future reference in the laboratory.

Precautions during collection and observation of specimens

1. Collect only the number of specimens needed to avoid wastage.
2. Do not harm the specimens during the collection exercise because it can distort the features of the specimens
3. Do not destroy the natural habitat of the of specimens.
4. Live specimens should be returned to their habitats whenever possible to maintain ecological balance.
5. Dangerous or injurious specimens (e.g. stinging insects or plants) should be handled with care (using forceps and gloves) to avoid injury/ for protection.
6. Highly mobile animals should be immobilized using suitable chemical substances (e.g. **Chloroform or diethylether)** for easy observation.

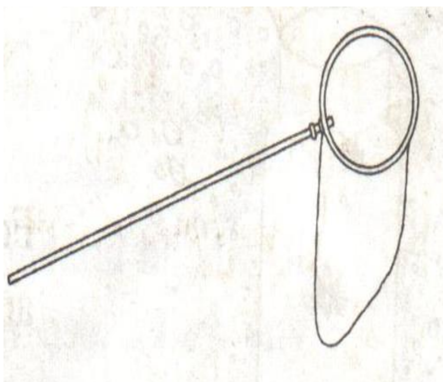
Sam obare

Apparatus used for collection of specimens

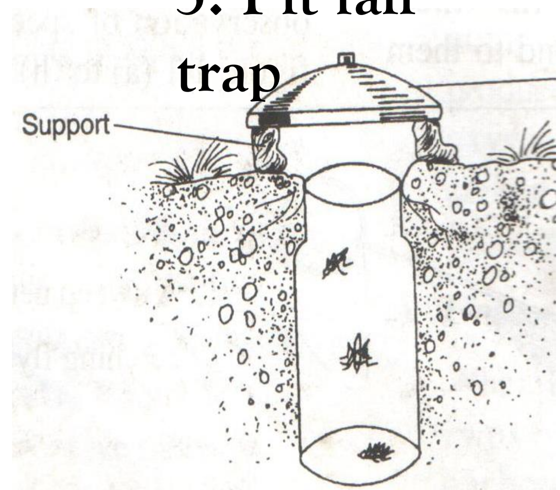
1. **Sweep net-** used to catch flying insects e.g. bees, butterflies, grasshoppers.
2. **Bait trap-**used for attracting and trapping small animals e.g. rats and mice.
3. **Pitfall trap-** it is used for catching crawling animals e.g. millipedes, spiders, ants, cockroach.
4. **Fish net-** used for trapping small fish and other water animals e.g. crabs and shrimps.
5. **Pooter-** it is used for sucking small animals from rock surfaces or barks of trees e.g. ants, termites.
6. **Pair of forceps-** it is used for picking up stinging animals and plants e.g. centipedes, spiders, stinging nettle.
7. **Specimen bottle-** it is required for keeping collected specimens.
8. **Hand lens-** it is used to enlarge objects and observe external features of collected specimens

Apparatus used for collection of specimens

1. Sweep net



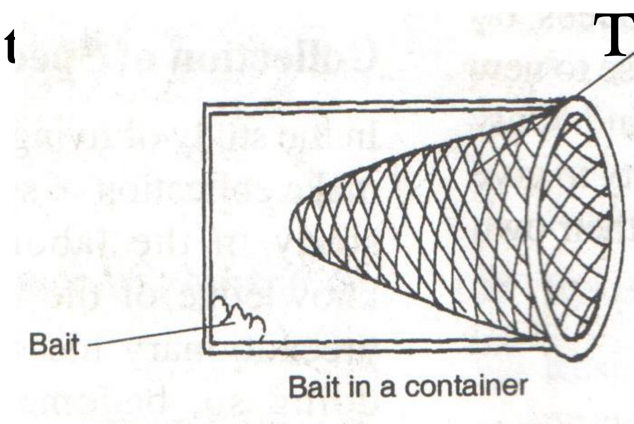
3. Pit fall trap



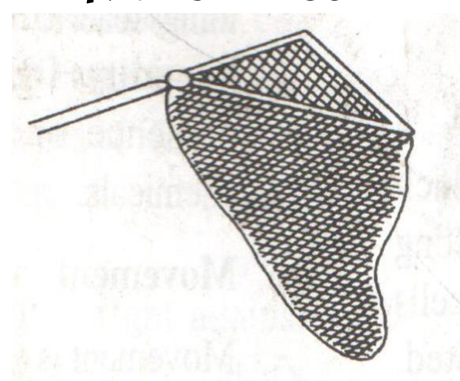
Cover to prevent rain from getting in

2. Bait t

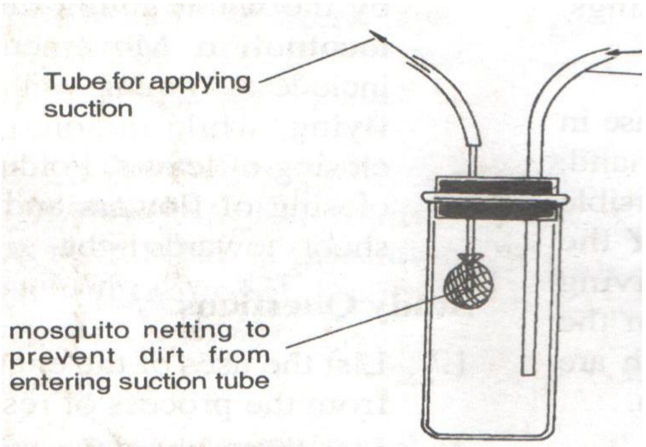
Trap



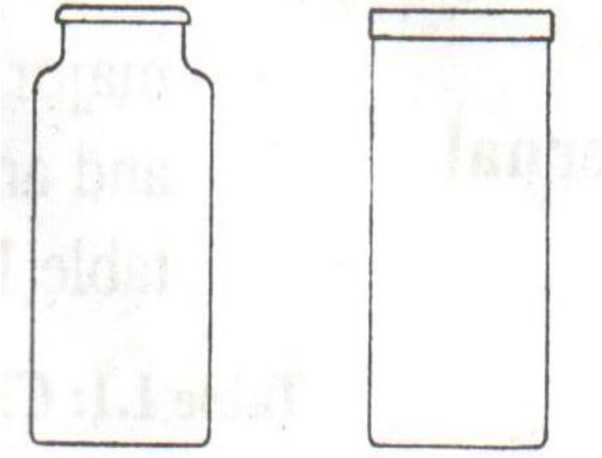
4. Fish net



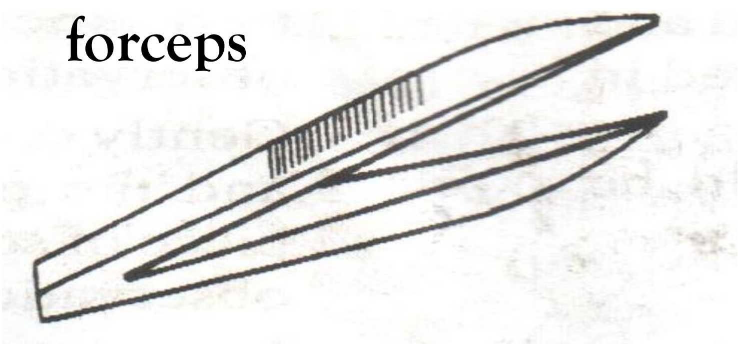
5. Pooter



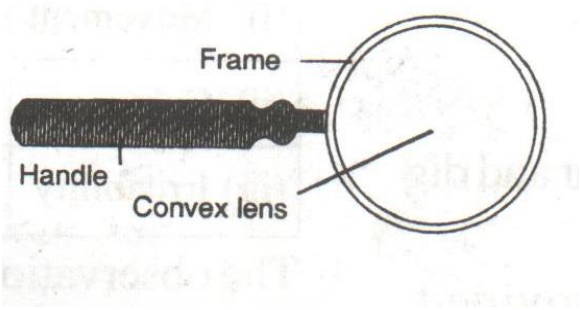
7. Specimen bottle



6. Pair of forceps



8. Hand lens



Differences between plants and animals

Plants	Animals
1. Most plants are green in colour/ have chlorophyll hence make their own food.	1. They lack chlorophyll hence do not make their own food/ feed on already manufactured food.
2. They respond to changes in their environment slowly.	2. They respond to changes in their environment faster.
3. They do not locomote/ move from one place to another.	3. They locomote/ move from one place to another.
4. Growth occurs at specific regions/ meristematic cells only.	4. Growth occurs all over the body.
5. They lack complex excretory and respiratory organs/ structures.	5. They have complex respiratory and excretory organs/ structures.